An Overview of Dedicated Hardware Designs for State-of-the-Art AV1 and H.266/VVC Video Codecs

Mário Saldanha[1], Marcel Corrêa[1,2], Guilherme Corrêa[1], Daniel Palomino[1], Marcelo Porto[1], Bruno Zatt[1], Luciano Agostini[1]
[1]Federal University of Pelotas – Graduate Program in Computing (PPGC) – Pelotas, Brazil
[2]Sul-Rio-Grandense Federal Institute – Bagé, Brazil
{mrdfsaldanha, mmcorrea, gcorrea, dpalomino, porto, zatt, agostini}@inf.ufpel.edu.br

***Abstract*—This paper presents an overview of published works related to dedicated hardware designs for the recent AV1 codec and the H.266/VVC video coding standard. We describe the main coding tools comparing them between AV1 and H.266/VVC codecs to highlight the new algorithms and techniques employed to improve coding efficiency. Then, we present the main published works proposing FPGA/ASIC hardware designs for AV1 and H.266/VVC. It is possible to observe that most of the published works focus on the challenge of implementing the new coding tools of both codecs to attend high throughput demands, such as real-time processing of 1080p, UHD 4K and UHD 8K resolutions.**

***Keywords*—AV1, VVC, video coding, hardware design**

I. Introduction

The continuous growth in consumption of digital videos over the internet, including services such as UHD (Ultra High Definition) and VR (Virtual Reality) streaming, as well as video sharing on social networks, are pushing the limits of available bandwidth of the telecommunication infrastructures. AOMedia Video 1 (AV1) [1] and H.266 Versatile Video Coding (VVC) [2] are the next generation video formats yet to be widely adopted to tackle these problems.

AV1 was released in June 2018 by the Alliance for Open Media (AOM) industry consortium, as the successor of VP9. It was developed with the main goal of being the state-of-the-art royalty-free video format, motivated by the fact that the major success of the internet is that its core technologies are open and freely implementable, and digital videos are a central part of the internet experience nowadays. The H.266/VVC standard was released in July 2020. It comes from a long line of successful video coding standards made by a joint effort of the Moving Picture Experts Group (MPEG) and the Video Coding Experts Group (VCEG) that includes its predecessors H.265 High Efficiency Video Coding (HEVC), H.264 Advanced Video Coding (AVC), and MPEG-2.

Even though next generations video codecs (encoder/decoder) such as AV1 and VVC are able to achieve a satisfactory rate-distortion performance for current video content, this efficiency comes at the cost of a high computational effort. This makes video coding an infeasible task for software solutions when real-time processing and high resolutions are desired, even when running on the most advanced general-purpose processors. Because of this, hardware acceleration has been applied to video coding. Although several hardware solutions were proposed to the predecessor standards, most of them cannot be used directly in AV1 and H.266/VVC, requiring new hardware designs.

This paper presents an overview of the hardware solutions related to AV1 and VVC published to date. Among these works, there are architectures for the modules of intra prediction, sub-pixel interpolation filters of the inter prediction, transforms, and in-loop filtering. The main focus of this paper is to provide an overview with content and references of related works focusing on hardware solutions for AV1 and H.266/VVC standards.

II. AV1 and H.266/VVC Tools and Comparisons

AV1 and H.266/VVC encoders follow a hybrid video coding scheme, which is based on the following signal and data processing operations: (i) inter and intra-frame prediction, (ii) de-correlating transform, (iii) quantization, and (iv) entropy coding. The prediction step is usually followed by the transform (T) and quantization (Q) of prediction residues, which is then followed by the entropy coding [3]. A reconstruction loop with inverse quantization (IQ) and inverse transform (IT) is also included to guarantee that the encoder and decoder will use the same reference data [3]. Finally, an In-loop Filter is also used to increase the subjective image quality [3]. The basic components of a generic hybrid video encoder are shown in Fig. 1. The solid black arrows indicate the data flow at the encoder from the input video sequence to the generated output bitstream.

Before the prediction process starts, a frame must be divided into several blocks of samples. Each video encoder defines a variable range of block sizes. The AV1 partition structure follows a 10-way partition tree and allows a total of 24 block sizes from the basic block (called superblock), which size can be 128x128 or 64x64, down to the minimum size 4x4, including asymmetrical blocks of 2:1 and 4:1 ratio [1]. H.266/VVC allows block sizes from 128x128 to 4x4 with a total of 28 block sizes from the basic block, including asymmetrical blocks obtained through binary and ternary split. The partition structure in H.266/VVC is called Quadtree with nested Multi-type Tree (QTMT) [2].

In intra-frame prediction, each block is predicted from the reconstructed samples of previously encoded spatial neighbor blocks from the same frame. AV1 allows the use of 56 directional modes and 13 non-directional intra prediction modes [1] whereas H.266/VVC allows the use of 65 directional modes, two non-directional and matrix-based intra prediction, which is the first learning-based tool in a video coding standard [2]. It is important to mention that the angles of the directional modes of both formats are not equivalent, nor are their interpolation filters.

In inter-frame prediction, the block is predicted from samples belonging to previously encoded frames. Both AV1 and H.266/VVC use Motion Estimation (ME) and Motion Compensation (MC) algorithms in this prediction. Motion vector prediction tools are used to reduce the amount of lateral

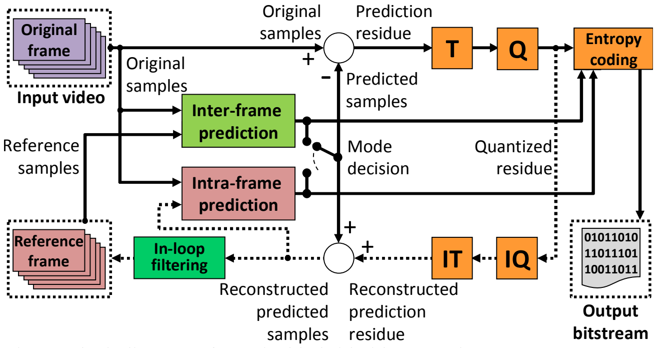

Fig. 1. Block diagram of a typical hybrid video encoder [3].

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data. Both codecs share tools like Skip (where no information is required to represent a block), Fractional Motion Estimation (FME) with 1/8 subsample precision in AV1 and 1/16 in H.266/VVC, and compound modes (where more than one prediction can be used in a single block). The main difference between the inter-prediction of AV1 and H.266/VVC are the algorithms used to implement the prediction tools.

In both intra and inter-frame prediction, the predicted and the original blocks are rarely equal, hence the difference between them must be calculated, encoded, and transmitted. This difference is called residue. The residue is processed by the transform module (T module, in Fig. 1), which converts the values from the spatial domain to the frequency domain in order to decorrelate the residue and concentrate the energy in fewer coefficients. Then, a quantization step (Q module in Fig. 1) is applied to the transformed coefficients to eliminate small values associated with spectral components that are not perceptually relevant, decreasing the amount of data to be encoded with controlled quality loss. H.266/VVC and AV1 use a variable number of transform types and sizes. AV1 uses four 1D transforms – the Discrete Cosine Transform (DCT), Asymmetrical Discrete Sine Transform (ADST), ADST applied in reverse order (flipADST), and Identity Transform (IDTX) – which can be combined resulting in 16 2D hybrid transforms [1]. The transforms are applied over 64x64 blocks and smaller, in a total of 19 different block sizes. H.266/VVC uses DCT-II, DCT-VIII, and DST-VII transforms, applied over 64x64 blocks and smaller, including rectangular shapes in a total of 25 different sizes. Besides, H.266/VVC applies Low Frequency non-Separable Transform (LFNST) to further decorrelate the transform coefficients of intra blocks [2].

The entropy coding processes the symbols (residual and lateral data) to reduce their statistical redundancy applying lossless algorithms. AV1 uses an adaptive multi-symbol arithmetic encoder [1] and H.266/VVC uses the Context-based Adaptive Binary Arithmetic Coder (CABAC) [2]. The two codecs also include in-loop filters to increase the subjective image quality, since the encoding process can generate undesired artifacts in the encoded video. AV1 employs the Constrained Directional Enhancement Filter (CDEF), Loop Restoration Filters, Frame Super-resolution, and Film Grain Synthesis [1], whereas H.266/VVC uses Luma Mapping with Chroma Scaling (LMCS), Deblocking Filter (DF), Sample Adaptive Offset filter (SAO), and Adaptive Loop Filter (ALF) [2].

Considering the high computational effort to encode and decode videos using AV1 and H.266/VVC codecs with all aforementioned tools enabled, dedicated hardware designs are mandatory, as previously discussed.

III. Related Works on AV1 Hardware Designs

Since the release of the AV1 bitstream specification, eight hardware related papers have been published. These works describe solutions for the modules of intra-frame prediction, inter-frame prediction, and in-loop filtering. Table I summarizes the results of these works. These early published works focused on new features of the AV1 that were not present in previous formats, such as the new non-directional intra prediction modes, the new angles of the directional intra modes, the new interpolation filters for the AV1 FME, and the novel CDEF in-loop filter.

Corrêa et al. [4] presented a non-directional intra prediction module for the encoder side, capable of processing a single prediction mode (Paeth) at very high speeds. This architecture covers all possible block partitions and reaches a throughput of UHD 8K at 30 frames per second (fps). A massive parallelism is used where a 32x32 block (or any smaller block) is processed in a single clock cycle.

Table I – Summary results of AV1 hardware designs.

Work	[4]	[5]	[6]	[7]
System	Encoder	Encoder	Encoder	Encoder
Module	Intra	Intra	Intra	Intra
Frequency	315MHz	648MHz	648MHz	1,296MHz
Technology	ASIC 40nm	ASIC 40nm	ASIC 40nm	ASIC 40nm
Area	247.3Kgates	109.6Kgates	128.5Kgates	455.8Kgates
Power	268.4mW	16.1mW	65.6mW	40.9mW
Throughput	7,680x4,320 @30fps	3,840x2,160@ 30fps	3,840x2,160@ 30fps	3,840x2,160 @60fps
Work	**[8]**	**[9]**	**[10]**	**[11]**
System	Encoder	Decoder	Decoder	Decoder
Module	Intra	Intra	Inter (FME)	Filter
Frequency	1,902MHz	132MHz	280Hz	23MHz
Technology	ASIC 40nm	ASIC 40nm	ASIC 40nm	ASIC 40nm
Area	691.7Kgates	89.4Kgates	141.1Kgates	369.9Kgates
Power	382.1mW	7.96mW	81.3mW	65.1mW
Throughput	3,840x2,160 @60fps	3,840x2,160 @60fps	7,680x4,320 @30fps	3,840x2,160 @60fps

Corrêa et al. [5] presented a non-directional intra prediction module for the encoder, capable of processing four intra prediction modes and all combinations of block sizes allowed by the AV1 specification. Similarly, Corrêa et al. [6] presented a non-directional intra prediction module for the encoder capable of processing 11 non-directional modes. As the number of supported modes and complexity of the architectures increased, the authors decided to decrease the parallelism and to optimize all multiplication blocks in order to keep the area and power within feasible limits. The parallelism strategy allowed the processing of one block row/column per clock cycle, thus the number of cycles depends on the width or height of the block, whichever is the largest. The authors in [5] reported a throughput enough to process UHD 4K at 30 fps. In [6], authors reported throughput to process UHD 4K at 30 fps.

Corrêa et al. [7] and Neto et al. [8] described directional intra prediction designs for the encoder that share similarities. Both designs support all 56 directional prediction modes and all block sizes, however only the design proposed in [8] support the four smoothing filtering processes of reference samples and the upsampling of reference samples. All 56 prediction modes are processed in parallel, but each predicted block is processed one row/column per clock cycle. Although the number of prediction modes being processed in parallel is quite large, a significant amount of redundant operations is reused in [7], because all predicted blocks share the same reference samples. This, however, cannot be done with the same degree of efficiency in the architecture proposed in [8], because for each of the 56 prediction modes, the encoder can use a different configuration of smoothing filtering and upsampling of the reference samples. Thus, the reference samples are not shared among predicted blocks. In [7], the authors reported a throughput of UHD 4K at 60 fps, while in [8] the authors reported a throughput of UHD 4K at 60 fps.

Goebel et al. [9] proposed a non-directional intra prediction module for the decoder that supports the DC and Chroma-from-Luma (CfL) prediction modes, all block sizes and 10-bits input samples. The sample level parallelism of the architecture allowed the processing of any block size as subblocks of size 4x4 (16 samples per cycle). In the decoding process, each block has to be predicted only once using the prediction mode signaled in the bitstream, hence the CfL unit of this design is only used for CfL coded blocks, but the DC unit is used for both modes, because the DC algorithm is one of the steps of the CfL prediction. Since a decoder must be able to deal with all features allowed by the AV1 specification, this design needs to support all other prediction modes to be a valid decoding design. The authors reported a throughput of UHD 4K at 60 fps.

Domanski et al. [10] presented an interpolation filter architecture for the MC step of the inter prediction module of the decoder. In AV1, a total of 90 different interpolation filters are used to obtain the interpolated subsamples, and the proposed architecture supports all of them. The sample level parallelism of the architecture allowed the processing of any block size as subblocks of size 4x4 (16 samples per cycle), but since it is a decoder design, only one of the many filters is used per predicted block (the one signaled in the bitstream). The authors reported a throughput of UHD 8K at 30 fps.

Zummach et al. [11] presented a CDEF architecture for the decoder. The CDEF process is applied to each area of size 8x8 within a frame, and the architecture was designed with enough parallelism to process an 8x8 area every three clock cycles. The architecture is composed of a direction search unit, which classifies the input texture with one of eight directions, and a filtering core unit, which filters the input texture using 64 filter kernels based on the detected direction. The authors reported a throughput of UHD 4K at 60 fps.

Commercial products for AV1 have been announced but most are under development phase. Known released hardware encoders are the Allegro AL-E210 [12] (UHD 4K) and the Dwango Real-Time Hardware Encoder [13] (HD 1,280x720). Known released decoders are the Realtek RTD1319 and RTD1311 [14] (UHD 4K) and the Broadcom BCM7218X [15] (UHD 4K). The manufacturers did not provide detailed architectural information regarding their products.

IV. Related Works on H.266/VVC Hardware Designs

There are a few works in the literature related to hardware architecture designs for the H.266/VVC encoder or decoder. In this section, we present works focusing on intra-frame prediction, fractional interpolation filters, and transform (encoder/decoder), considering FPGA and ASIC-based designs. Table II summarizes the works discussed in this section. These works focused on the novel H.266/VVC tools such as the new intra prediction modes, the new fractional interpolation filters, and the new transform set. Besides, the proposed solutions explored advanced techniques to provide high throughput (at least 1,920x1,080 at 30fps) with low hardware resource usage and low power dissipation.

Azgin et al. [16] proposed a hardware architecture for the H.266/VVC intra prediction encoder, considering the 65 angular prediction modes and square-shaped blocks with sizes of 4x4, 8x8, 16x16, and 32x32. Since there are several angular prediction equations of cubic and gaussian filters to be evaluated for different block sizes, a strategy of data reuse is employed to calculate identical prediction equations only once. This strategy explores identical prediction equations that are used in prediction modes for the same or different block sizes. Moreover, the equations are calculated using DSP blocks targeting low-energy consumption and faster multiplication results. The hardware design was synthesized for a Xilinx Virtex-7 FPGA and can process 1,920x1,080 videos at 34 fps. However, the proposed solution is not capable of processing non-angular prediction modes, rectangular-shaped blocks, and 64x64 blocks, limiting the coding efficiency.

Table II – Summary results of H.266/VVC hardware designs.

Work	[16]	[19]	[27]	[28]
System	Encoder	Encoder	Encoder	Decoder
Module	Intra	Inter (FME)	Transform	Transform
Frequency	119 MHz	227 MHz	250 MHz	600 MHz
Technology	FPGA 28nm	FPGA 28nm	ASIC 65nm	ASIC 28nm
Area	46.4 KLUTs	9.3 KLUTs	496.4Kgates	96.9Kgates
Power	–	320.18mW	62.6mW	–
Throughput	1,920x1,080 @34fps	1,920x1,080 @47fps	7,680x4,320 @160fps	3,840x2,160 @30fps

Mert et al. [17] and Azgin et al. [18][19] presented hardware architectures for H.266/VVC fractional interpolation filters. The works [17] and [18] implement the original fractional interpolation filters without simplifications. In [19], an optimized version of these implementations was proposed. This solution employs an approximation of the original H.266/VVC filters coefficients, which considers that neighboring samples have similar values due to the spatial correlation, in this case, small coefficient values have little influence in the filter results. Thus, this solution designed 14 different 3-tap filters and one 4-tap filter instead of 15 8-tap filters defined in the H.266/VVC. The proposed approximate H.266/VVC fractional interpolation filters were implemented in the VVC Test Model (VTM) and presented a BD-rate increase of up to 0.52%. The synthesis results were generated targeting Xilinx Virtex-7 FPGA and demonstrated that the hardware architecture can process 1,920x1,080 videos at 47 fps. A comparison using the original 15 8-tap filters defined in the H.266/VVC showed that the proposed solution reduces power dissipation by up to 40%.

The works [20]-[28] presented hardware architectures for the H.266/VVC transform module. These works proposed different approaches to implement forward/inverse transforms. Due to space limitation we chose to discuss the works [27] and [28], which proposed hardware solutions to encoder and decoder, respectively.

Fan et al. [27] developed a high-performance pipelined hardware implementation for 2D DST-VII/DCT-VIII transforms operations in H.266/VVC, considering square and rectangular-shaped block sizes. The hardware architecture consists of three main modules: (i) 1D row transform, (ii) 1D column transform, and (iii) transpose memory. The modules (i) and (ii) have the same structure, only with different bit-width parameters. The 1D transform modules were designed with Shift-Addition Units (SAUs) to perform the matrices multiplications. Additionally, the SAUs were designed by employing the N-Dimensional Reduced Adder Graph (RAG-n) algorithm. Since DST-VII/DCT-VIII transforms have the same coefficients in each row but in inverse order, the hardware benefits from DST-VII architecture to implement the DCT-VIII by only inverting the order of the inputs and assigning appropriate outputs signs. To implement the transpose memory a dual-port SRAM is employed since the pipelined architecture needs to read and write simultaneously. To achieve the throughput rate of 32 samples per cycle, the transpose memory is divided into 32 banks using a diagonal scheme. To calculate different block sizes, the proposed architecture considers that all sizes are within a 32x32 block size. The hardware design was capable of processing 7,680x4,320 videos at 160fps. When the proposed solution is compared with an approach using multipliers and register array, the area and power dissipation are reduced by 47% and 10%, respectively.

Farhat et al. [28] proposed a hardware design for the transform module of the H.266/VVC decoder. The proposed solution supports 1D Inverse DCT-II (IDCT-II) and IDST-VII/IDCT-VIII with sizes ranging from 4 to 64 and from 4 to 32, respectively. The 2D transform process is performed using two 1D transform modules by adding an intermediate transpose memory. Thus, this solution is capable of processing all MTS transform cores and all block sizes (including square and rectangular blocks). The hardware design has a fixed throughput of 2 samples per cycle and a fixed latency for all block sizes based on the largest block latency to provide a fully pipelined structure. The authors proposed two designs of the hardware architecture: (i) using adders and shifts instead of multipliers (MCM) and (ii) using regular multipliers (RM). Synthesis results demonstrated that the architecture using RM consumed 63% fewer gates than MCM, being capable of processing 3,840x1,260 videos at 30fps.

V. CONCLUSIONS

In this paper we presented an overview of the main works published to date in the literature that focus on hardware designs for the AV1 and H.266/VVC video codecs. An analysis over eight and thirteen works focusing on AV1 and H.266/VVC, respectively, shows that the main challenge consists of implementing the newly introduced coding tools of both codecs for high-throughput demands, such as real-time processing of 1080p, UHD 4K and UHD 8K resolutions. Considering AV1, the existing hardware solutions focus mainly on intra-frame prediction, inter-frame prediction, and in-loop filtering operations. However, there are still no hardware-based solutions published to date for any of the 16 combinations of 2D hybrid transforms allowed in AV1, which is one of the main novelties introduced in the codec. Regarding H.266/VVC, the works focus on the intra-frame prediction, fractional interpolation filters, and transform modules. Still, there are currently no available solutions for inter-frame prediction, typically the most computationally intensive module of modern codecs. In conclusion, there are several challenges and a wide space for novel contributions focusing on efficient high-throughput hardware design for the emerging AV1 and H.266/VVC codecs.

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